

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 1
timesteps_orig: 30
timesteps: 30
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 2

LSTM Multi-Step Training Report

Report Generated: 2025-05-09 11:12:29

Training Start: 2025-05-09_10-02-09

Training End: 2025-05-09_10-32-19

Training Duration: 30.16 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 0.6113

MSE: 0.7159

RMSE: 0.8461

R2 Score: 0.9606

Multi-Step Forecast:

Forecast Steps: 2

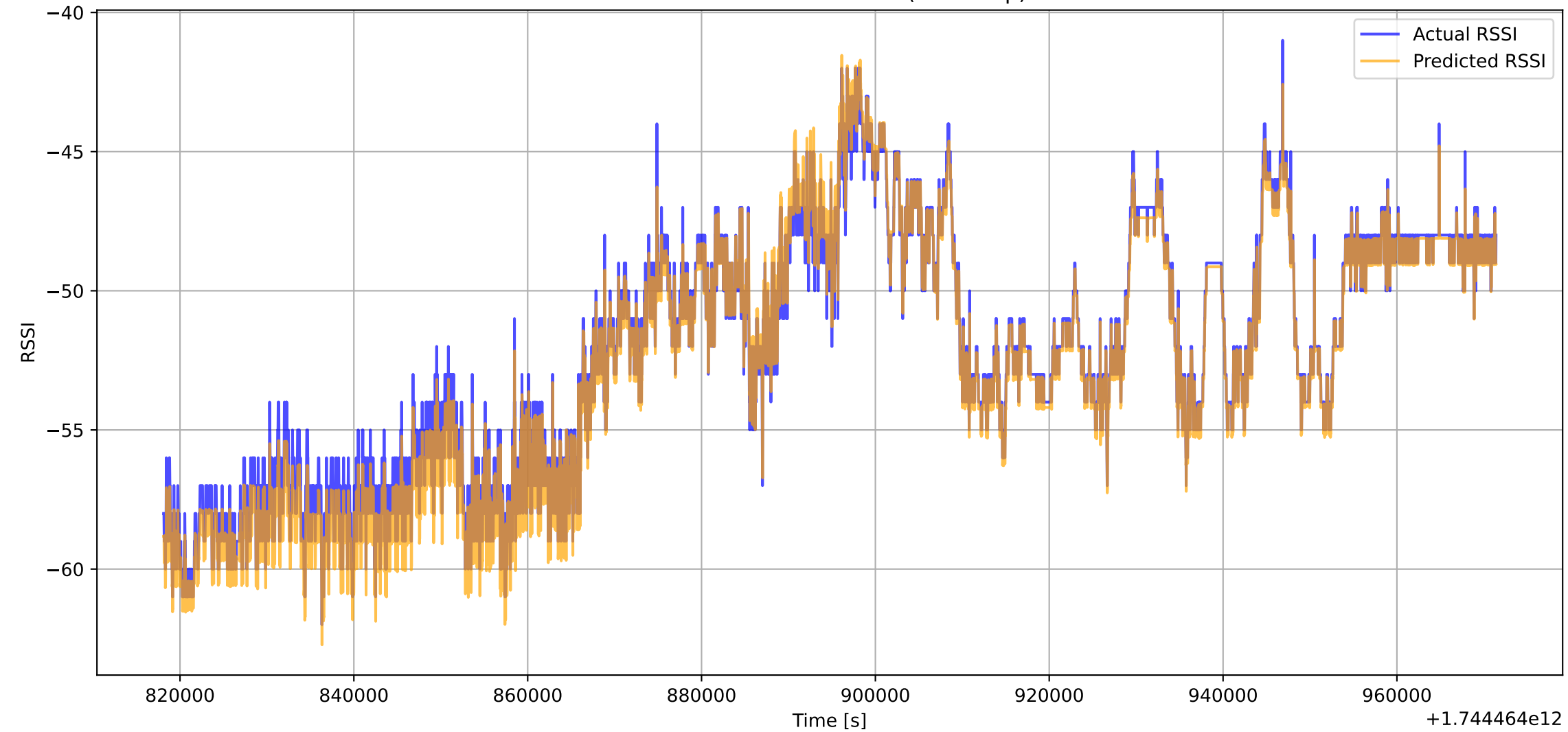
MAE: 0.0084

MSE: 0.0001

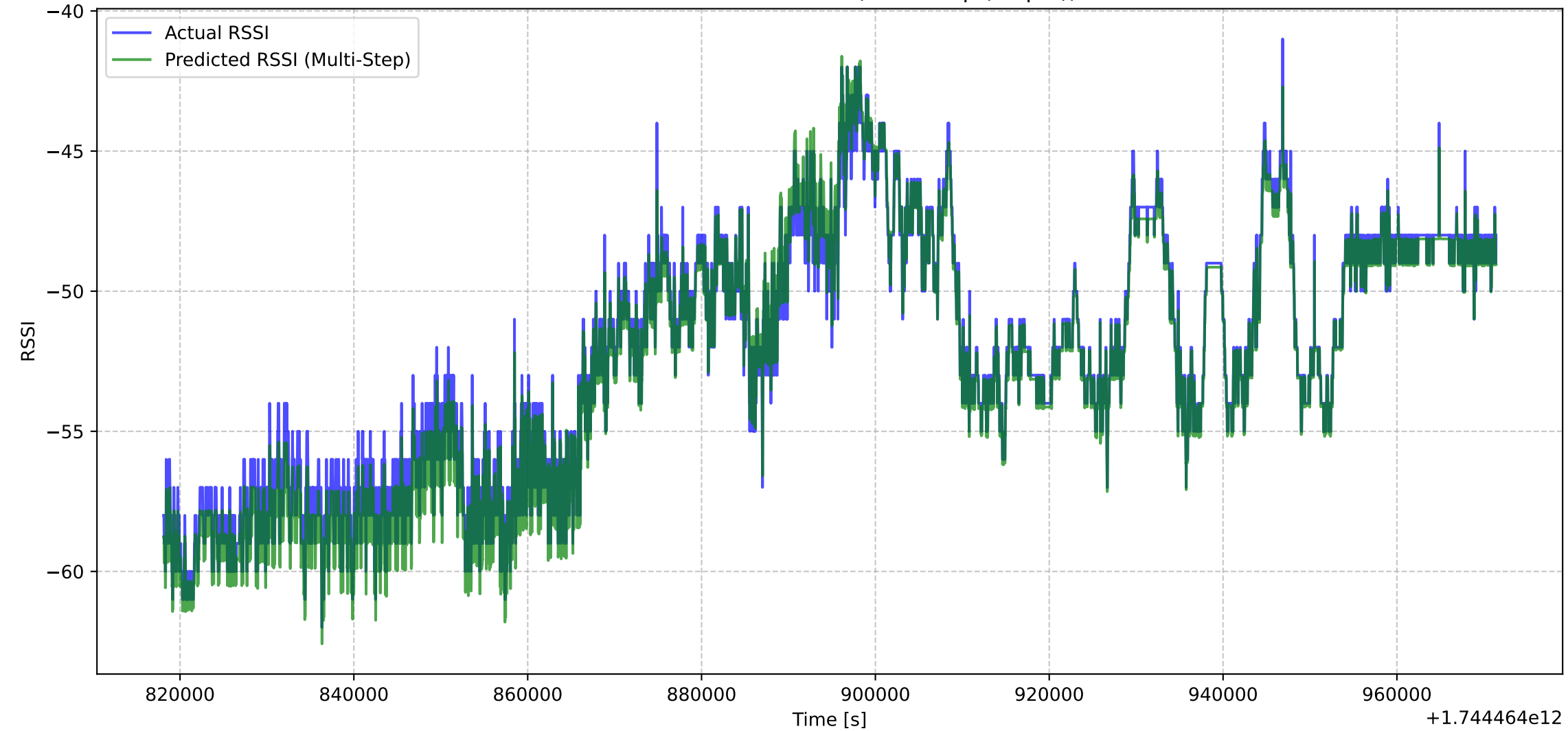
RMSE: 0.0116

R2 Score: 0.9668

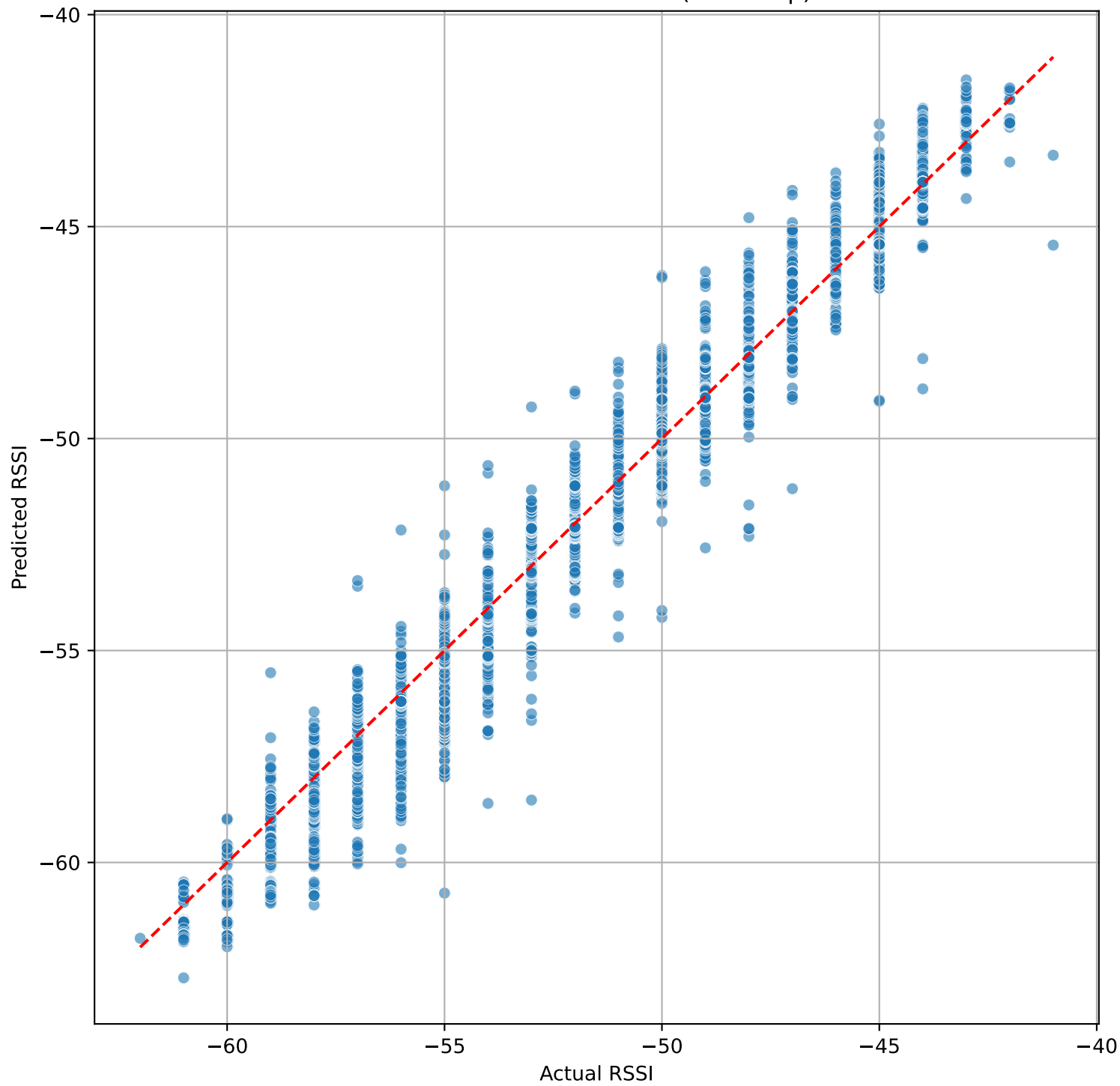
Actual vs Predicted RSSI (First Step)



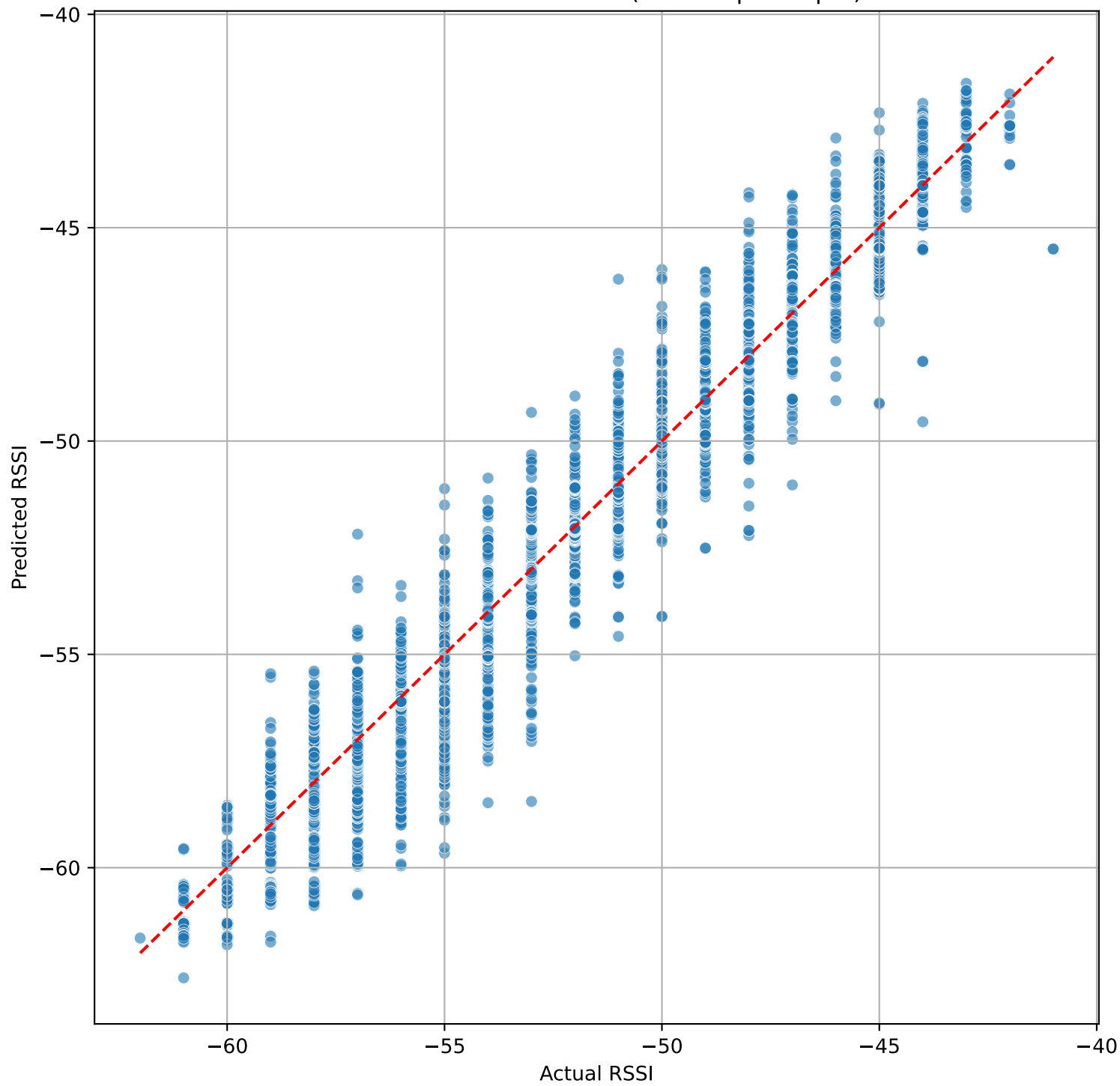
Actual vs Predicted RSSI (Multi-Step (step 2))



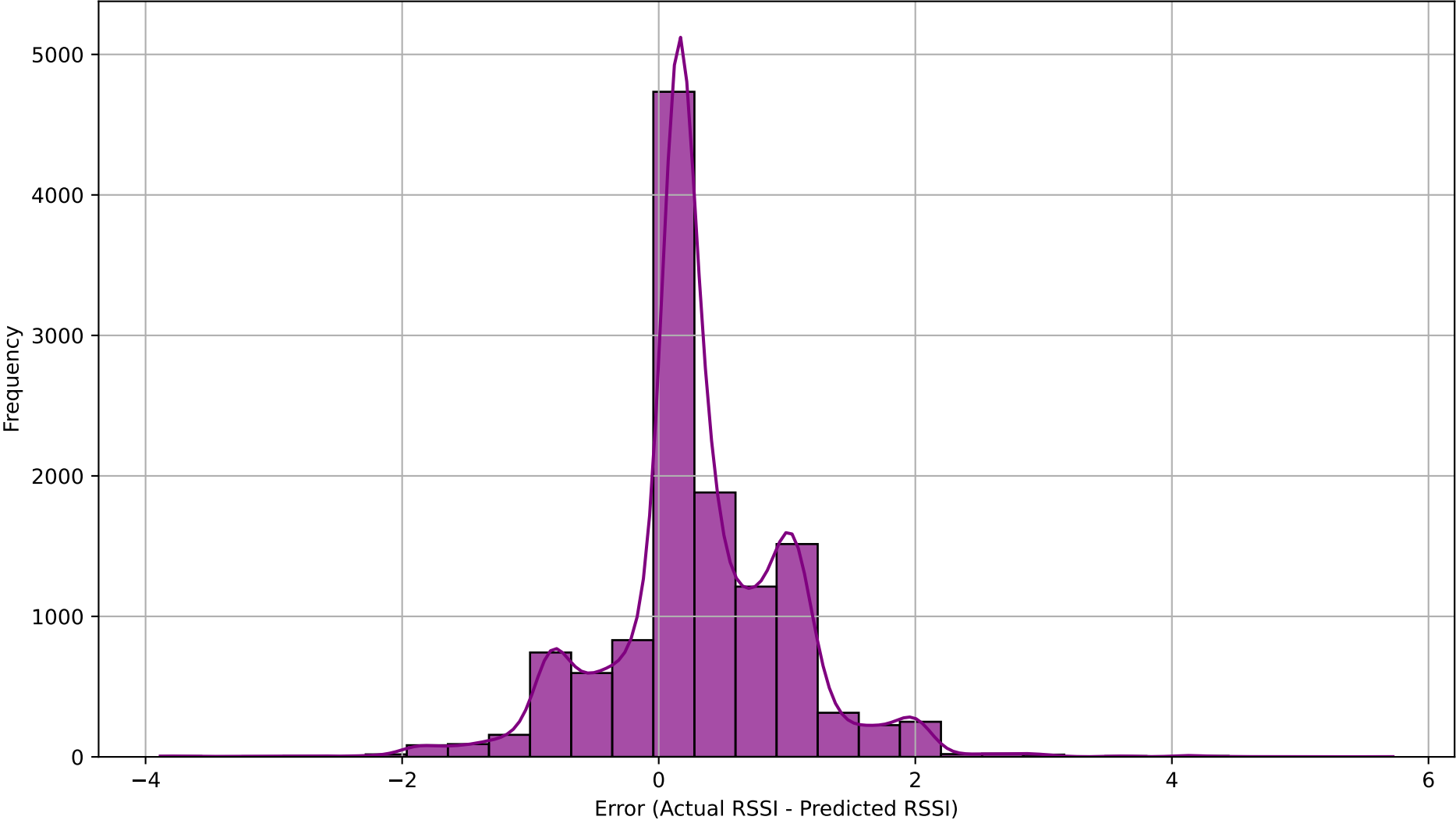
Actual vs Predicted RSSI (First Step)



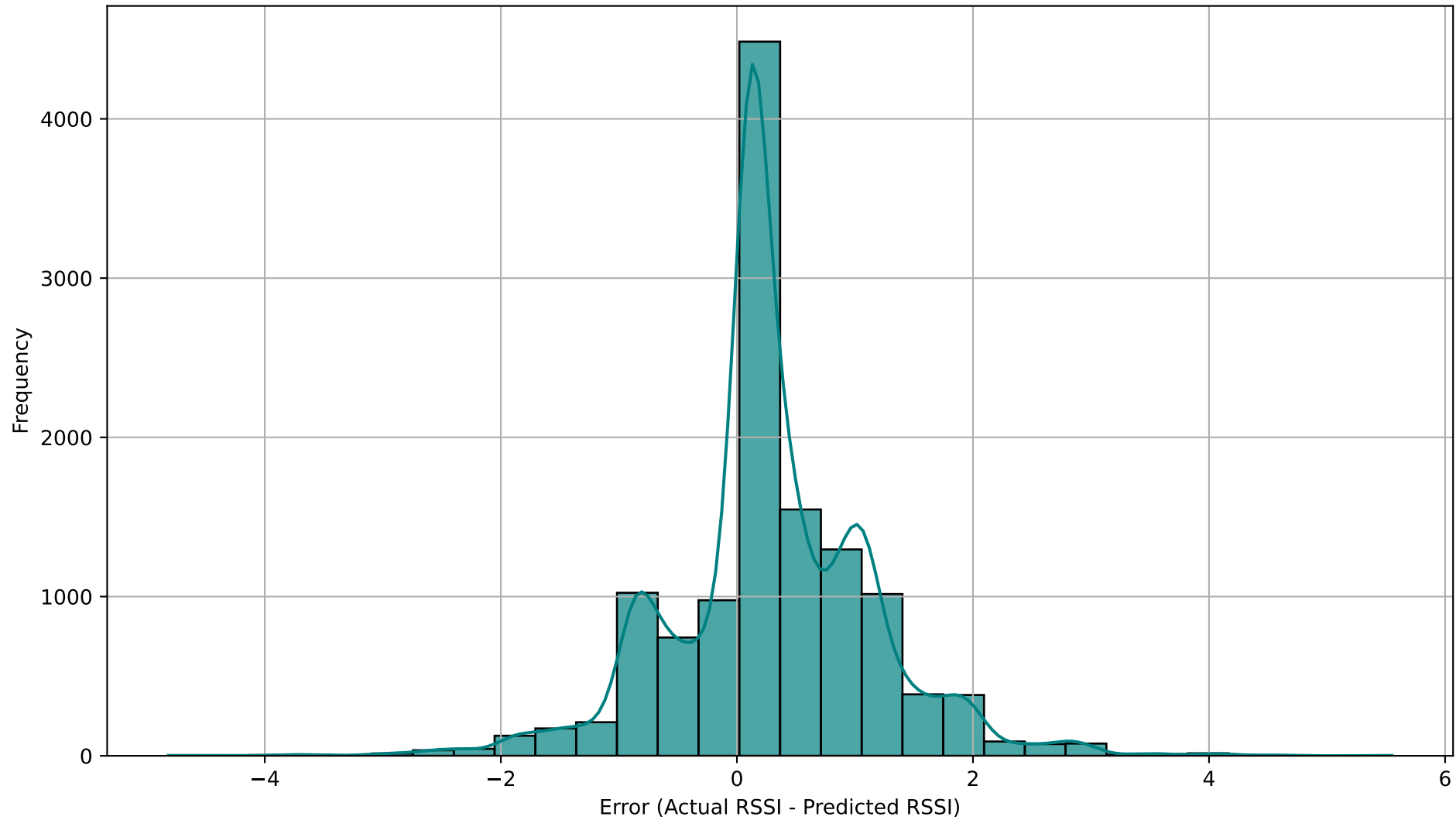
Actual vs Predicted RSSI (Last Step - step 2)



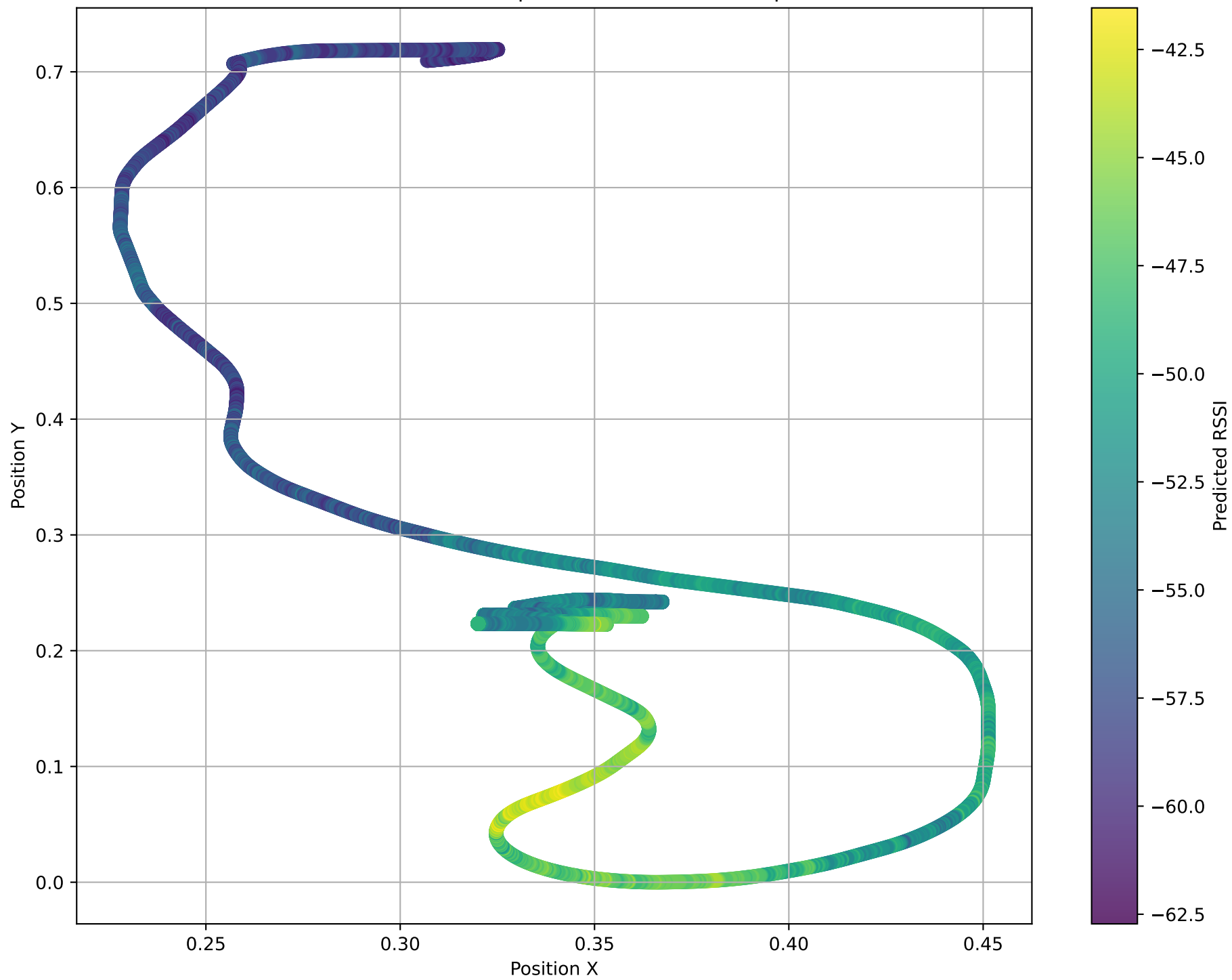
Distribution of Prediction Errors (First Step)



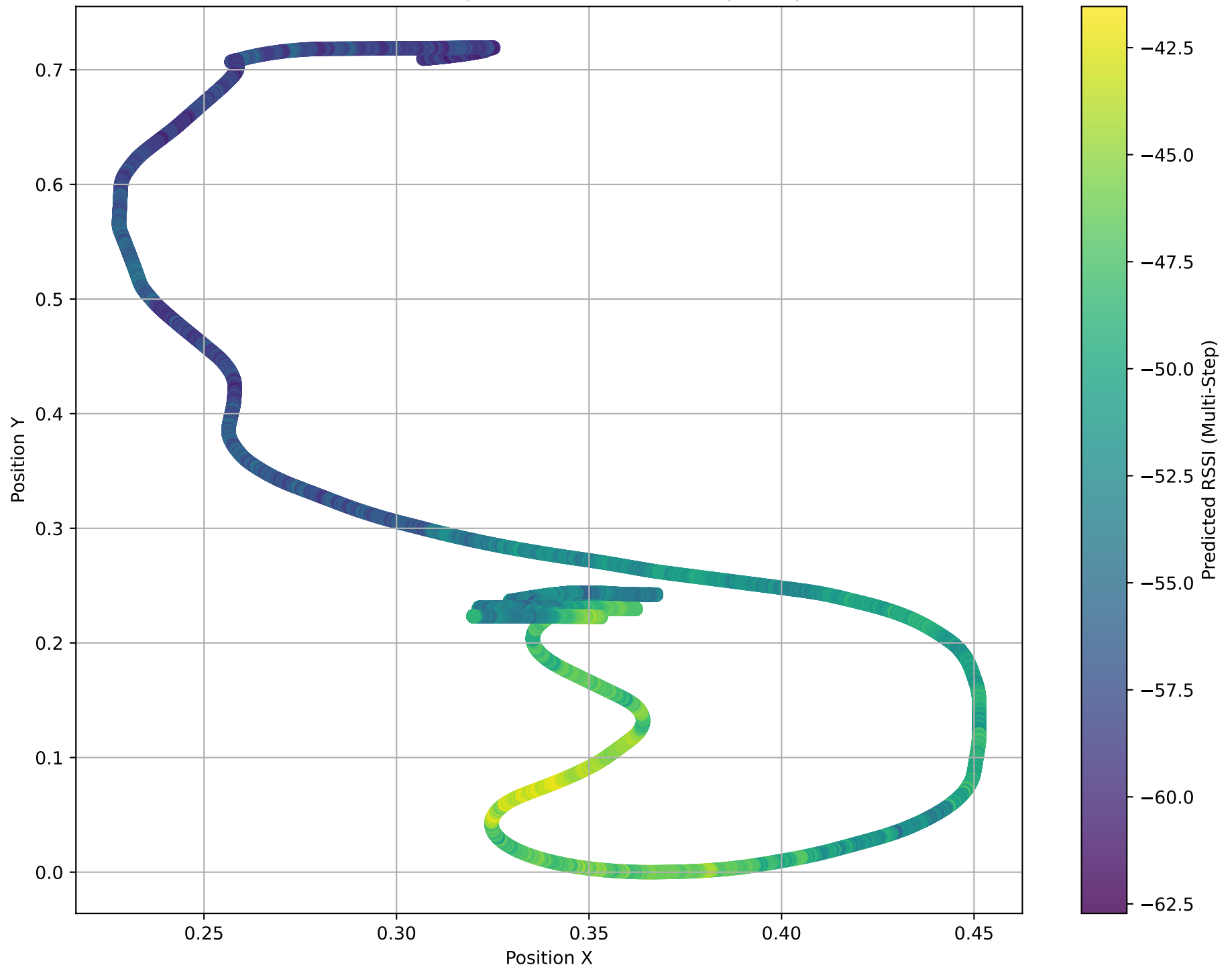
Distribution of Prediction Errors (Last Step - step 2)



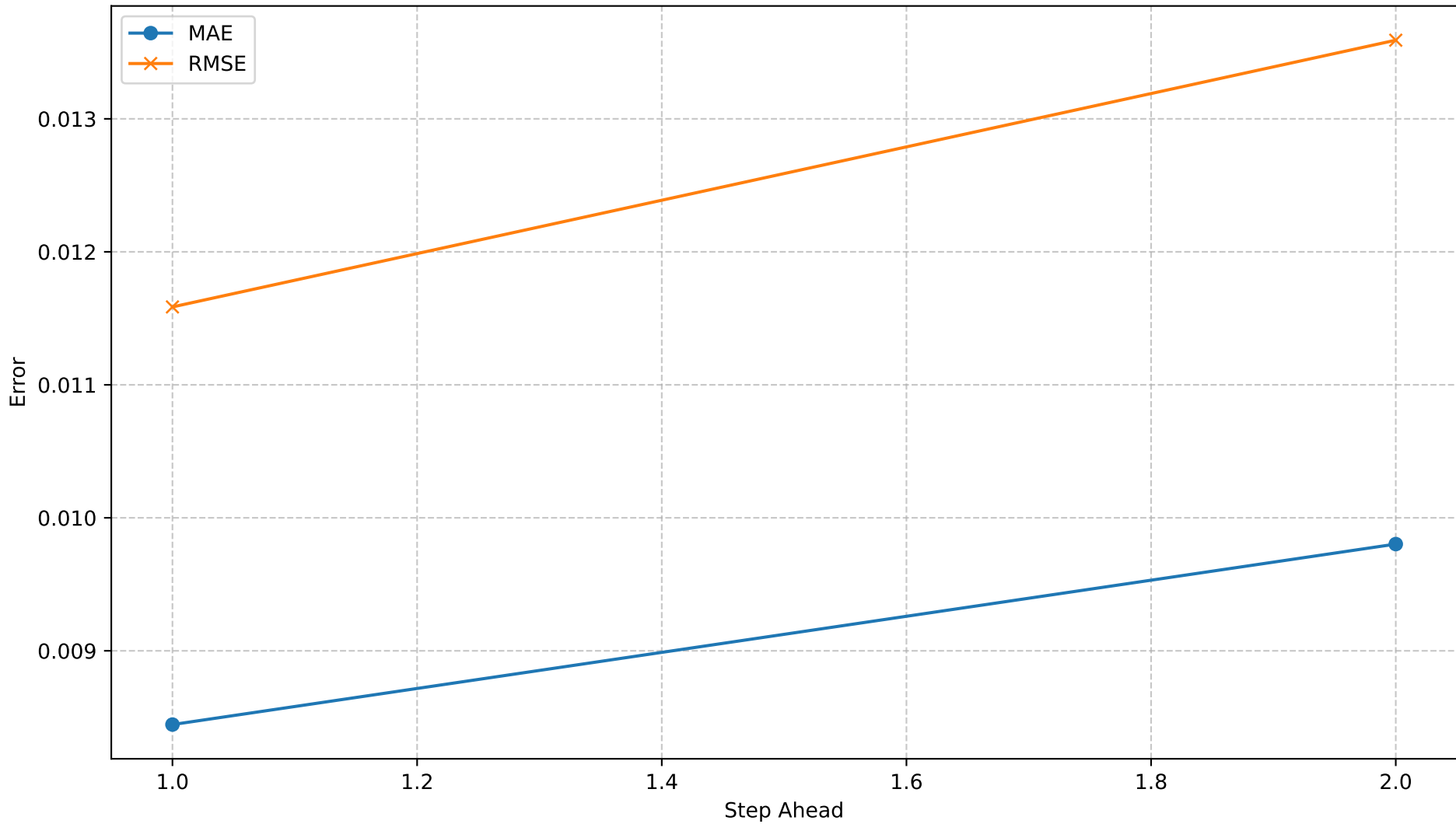
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 2)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 2

MAE / RMSE (dB)

- 1º passo : 0.01 / 0.01
- Mediana : 0.01 / 0.01
- 2º passo : 0.01 / 0.01

Crescimento do erro: 16.1% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 2 (0.01), 1 (0.01)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.